

EXPERIMENT NO 2

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#include <iostream>

#include <queue>

#include <unordered_map>

#include <vector>

using namespace std;

// Node structure for Huffman Tree

struct Node {

    char ch;

    int freq;

    Node *left, *right;

    Node(char c, int f) : ch(c), freq(f), left(nullptr), right(nullptr) {}

};

// Comparator for priority queue (min-heap)

struct Compare {

    bool operator()(Node* l, Node* r) {

        return l->freq > r->freq; // Min-heap based on frequency

    }

};

// Recursive function to generate Huffman Codes

void generateCodes(Node* root, string str, unordered_map<char, string>& huffmanCode) {

    if (!root)

        return;

    // Leaf node contains character

    if (!root->left && !root->right) {

        huffmanCode[root->ch] = str;

    }

    generateCodes(root->left, str + "0", huffmanCode);

    generateCodes(root->right, str + "1", huffmanCode);

}
```

```

int main() {
    string text;

    cout << "Enter the string to encode: ";
    getline(cin, text);

    // Count frequency of each character
    unordered_map<char, int> freq;
    for (char ch : text)
        freq[ch]++;

    // Create a priority queue (min-heap) of nodes
    priority_queue<Node*, vector<Node*>, Compare> pq;

    for (auto pair : freq) {
        pq.push(new Node(pair.first, pair.second));
    }

    // Build Huffman Tree
    while (pq.size() != 1) {
        Node* left = pq.top(); pq.pop();
        Node* right = pq.top(); pq.pop();

        Node* sum = new Node('\0', left->freq + right->freq);
        sum->left = left;
        sum->right = right;
        pq.push(sum);
    }

    Node* root = pq.top();

    // Generate Huffman Codes
    unordered_map<char, string> huffmanCode;
    generateCodes(root, "", huffmanCode);

    // Print Huffman Codes

```

```

cout << "\nHuffman Codes:\n";
for (auto pair : huffmanCode)
    cout << pair.first << " : " << pair.second << "\n";

// Encode the input string
string encoded = "";
for (char ch : text)
    encoded += huffmanCode[ch];

cout << "\nEncoded string:\n" << encoded << endl;

return 0;
}

```

OUTPUT

icoer@icoer-Vostro-3470:~\$ g++ huffman.cpp

icoer@icoer-Vostro-3470:~\$./a.out

Enter the string to encode: shravani

Huffman Codes:

r : 111

s : 011

h : 010

a : 10

n : 001

v : 110

i : 000

Encoded string:

0110101111011010001000